

TYPES OF CANCER

Bone cancer



BONE CANCER BOOKLET: EVERYTHING YOU NEED TO KNOW

What is bone cancer?

Bone cancer can refer to primary bone cancer or [secondary bone cancer](#) and the two types are quite different. Primary bone cancer is cancer that begins in the bones. Secondary (metastatic) bone cancer refers to a cancer that started elsewhere in the body and has spread to the bones.

- Symptoms
- Diagnosis
- Treatment

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Understood



Primary bone cancer is rare. It is estimated that 269 people were diagnosed with bone cancer in 2024. The average age at diagnosis is 42 years old. More commonly diagnosed is secondary bone cancer.

If you have secondary cancer in the bone, it may be useful to read information about the primary cancer in conjunction with this information, as secondary bone cancer is treated differently to primary bone cancer. A secondary cancer is named after the primary site where it began, for example lung cancer with bone secondaries.

There are more than 30 different types of primary bone cancer. Also called bone sarcoma, the most common types include:

osteosarcoma, which starts in cells that grow bone tissue

chondrosarcoma, which starts in cells that grow cartilage

Ewing's sarcoma, which affects cells in the bone or soft tissue that multiply rapidly.

Some types of cancer affect the soft tissue around the bones and these are known as [soft tissue sarcomas](#) and may be treated differently.

The information below refers to primary bone cancer.

► **PREVENTION**

► **PROGNOSIS**

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## Bone cancer signs and symptoms

The most common symptom of bone cancer is pain in the bones and joints, which

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stiffness or tenderness in the bone

problems with movement

unexplained weight loss

a fractured bone

loss of feeling in the affected limb

tiredness

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## Causes of bone cancer

Some factors that can increase your risk of bone cancer include:

previous radiotherapy, particularly for people who received high doses at a young age

other bone conditions, such as Paget's disease of the bone

genetic factors, such as inherited conditions like Li-Fraumeni syndrome, and a strong family history of certain cancers.

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## Blood tests

Blood tests will include a full blood count to check your general health.

## X-rays

X-rays are used to scan the bones to reveal any damage.

## CT scan

A **CT (computerised tomography)** scan is a test using x-rays and a computer to create detailed pictures of the inside of your body. The computer will then put the images together to create a three dimensional image that can highlight any bone abnormality.

## MRI

MRI scans create pictures of your body that help highlight any bone abnormality.

## Biopsy

The removal of some cells and tissues from the affected bone for examination under a microscope.

## After a diagnosis of bone cancer

After finding out you have primary bone cancer, you may feel shocked, upset, anxious or confused. These are normal responses. Talk to your doctor about what you can

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## Treatment for bone cancer

Primary bone cancer is usually treated with [surgery](#), [chemotherapy](#), [radiotherapy](#) or a combination of these treatments.

### Staging

If cancer is detected, your doctor will determine the grade and stage of your cancer. Grading refers to how quickly a tumour might grow. Staging describes how far cancer has spread.

### Limb-sparing surgery

Surgery to remove the cancer but spare the limb is the most common type of surgery for bone cancer. Under general anaesthetic, the surgeon removes the affected part of the bone where the cancer is growing. The bone that is removed is replaced with either an implant (prosthesis) or a bone graft, using a healthy part of bone from another part of the body or from a "bone bank".

### Amputation

When it is not possible to remove all of the cancer without affecting the arm or leg too much, the limb will be removed. This is becoming increasingly less common. After

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Surgery usually involves removing the cancer, the affected bone along with some healthy tissue around it. Bones from other parts of the body may be used to replace the bone that was removed.

## Chemotherapy

Chemotherapy drugs are used to destroy or damage cancer cells. For certain types of bone cancer, chemotherapy can be used in combination with surgery to either shrink the tumour before surgery, or after surgery to kill any cancer cells left behind.

Chemotherapy can also be used as a form of palliative treatment, to arrest the growth, or control the symptoms of advanced cancer.

## Radiation therapy (radiotherapy)

Radiation therapy uses high energy rays to destroy or damage cancer cells. It may be given before surgery, to help make surgery easier and safer, or after surgery, to reduce the chance of the cancer cells regrowing.

## Palliative care

In some cases of bone cancer, your medical team may talk to you about [palliative care](#). Palliative care aims to improve your quality of life by alleviating symptoms of cancer. As well as slowing the spread of bone cancer, palliative treatment can relieve pain and help manage other symptoms. Treatment may include radiotherapy, chemotherapy or other drug therapies.

## Treatment Team

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**Understood**

GP (General Practitioner) - looks after your general health and works with your specialists to coordinate treatment.

Surgeon - surgically removes tumours and performs some biopsies.

Radiation oncologist - prescribes and coordinates radiation therapy treatment.

Medical oncologist - prescribes and coordinates the course of chemotherapy.

Pain management team - treats pain, particularly if it is difficult to control or severe.

Cancer nurse - assists with treatment and provides information and support throughout your treatment.

Other allied health professionals - such as social workers, pharmacists, and counsellors

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## Screening for bone cancer

There is currently no national screening program for bone cancer available in Australia.

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## Preventing bone cancer

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## Prognosis for bone cancer

It is not possible for a doctor to predict the exact course of a disease, as it will depend on each person's individual circumstances. However, your doctor may give you a prognosis, the likely outcome of the disease, based on the type of bone cancer you have, the test results, the rate of tumour growth, as well as your age, fitness and medical history.

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## Sources

*Understanding Primary Bone Cancer information fact sheet*, Cancer Council Australia © 2023. Last medical review of source fact sheet: June 2019. This information was reviewed by: Prof Peter Choong AO, Orthopaedic Surgeon, and Sir Hugh Devine Professor, St Vincent's Hospital, and Head of Department of Surgery, The University of Melbourne, VIC; Catherine Chapman, Adolescent and Young Adult and Sarcoma Cancer Specialist Nurse, Division of Cancer and Ambulatory Support, Canberra Hospital, ACT; A/Prof Paul Craft AM, Medical Oncologist, Canberra Hospital and Australian National University, ACT; Belinda Fowlie, Bone Tumour Nurse Practitioner Candidate, SA Bone and Soft Tissue Tumour Unit, Flinders Medical Centre, SA; Prof Angela Hong, Radiation Oncologist, Chris O'Brien Lifehouse, and Clinical Professor, The University of

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